GOAL IN ONE

LEVELS 41-60

20 authored physics-puzzle concepts

Twelve compact arenas, eight scrolling worlds, purpose-built obstacles, camera annotations and machine-readable geometry for Codex.

ILLUSTRATED DESIGN DOCUMENT + JSON ENGINEERING DATA

CURRENT CODE AUDIT: 26 OCT 2026 | NO CODE CHANGES MADE



EXAMPLE / LEVEL 55 - THREE CHAMBERS

Designer diagrams describe geometry and intended routes. Physical feasibility still requires Codex playtests.

DESIGN STATUS: STRUCTURAL CHECKS PASS. NO ENGINE PLAYTEST CLAIMED.

THE DESIGN SYSTEM

EXISTING ENGINE CONTRACT / VISUAL LEGEND / REVIEW TARGETS

Verified against repository code

Latest observed commit: f1aa612f (26 Oct 2026). The code currently builds 40 levels via levels.ts, extra-layouts.ts and expansion.ts.

Ball radius 16; wall thickness 10.4; fixed timestep 120 Hz. Existing mechanics: springs, winds, breakables, gates, switches, rotors, slippery ice, moving blocks and floor boost/delete pads.

Goal detection requires the football to cross the right-facing mouth plane, not just approach the goal sprite.

Engine changes needed before scrolling levels

Add per-level camera properties and calculate real world bounds rather than always fitting the fixed 1,250-unit width.

Camera follows ball smoothly in x (and y for tall worlds), with dead zone, zoom floor, clamping, and fully camera-aware pointer drag aiming.

Keep all 40 existing levels, their save data and current mechanics intact. Make goal net part of visible right boundary.

VISUAL SYMBOLS

- WOODEN SOLID OBSTACLE
- TIMBER ARENA FRAME
- SWITCH / GATE LINK
- SPRING PAD
- UPDRAFT / WIND
- LOW-FRICTION ICE
- BREAKABLE WOOD
- RED RESET HAZARD
- INTENDED BALL ROUTE
- SCROLL CAMERA WINDOW
- GREEN GOAL

The green route is a proposed player strategy, not a simulated football path. Straight segments deliberately simplify rebounds.

LEVEL CATALOGUE / 41-60

12 COMPACT + 8 SCROLLING / 20 DISTINCT PUZZLE IDEAS

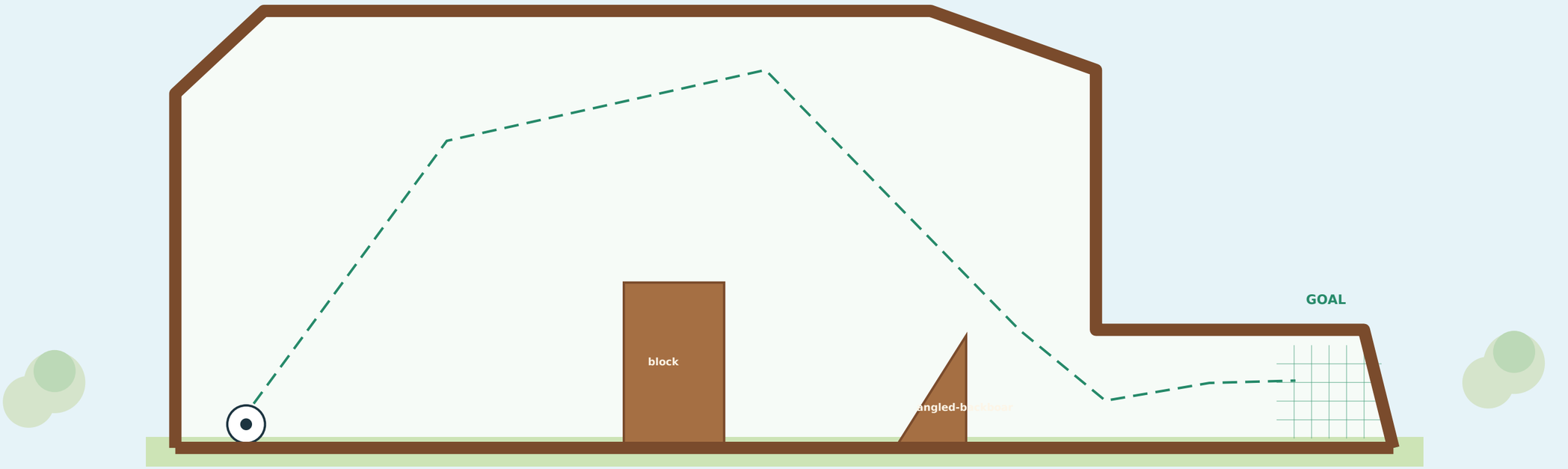
41	**Roof Ricochet**	banks / ramp	FIXED	51	**High Switch**	switch / height-change / required-gate	FIXED
42	**Window of Opportunity**	partition / precision	FIXED	52	**The Z-Corridor**	zigzag / bank	FIXED
43	**Unlock the Passage**	switch / required-gate	FIXED	53	**Longshot Relay**	scroll-x / spring / rotor	SCROLL
44	**Springboard**	spring / arc	FIXED	54	**Updraft Shaft**	scroll-xy / updraft / upper-doorway	SCROLL
45	**Above or Below**	two-routes / raised-platform	FIXED	55	**Three Chambers**	scroll-x / sequential-switches / required-gates	SCROLL
46	**Crosswind Slot**	wind / aerial-route	FIXED	56	**Hazard Overpass**	scroll-x / bridge / hazard	SCROLL
47	**Break or Bypass**	breakable / optional-shortcut	FIXED	57	**Wind and Workshop**	scroll-x / wind / required-gate	SCROLL
48	**Clockwork Gap**	rotor / timing	FIXED	58	**Skybridge Return**	scroll-xy / high-gate-gap / skybridge	SCROLL
49	**Ice Corridor**	ice / momentum	FIXED	59	**Switchback Factory**	scroll-x / sequential-switches / ice	SCROLL
50	**The Timed Door**	timed-gate / switch	FIXED	60	**The Grand Route**	scroll-x / combined / required-gate	SCROLL

41 Roof Ricochet

SINGLE-SCREEN • DIFFICULTY 2/5 • SUGGESTED OPTIMAL 1 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED | BANKS | RAMP

41 / ROOF RICOCHET



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

A deliberate bank shot off a level ceiling and angled backboard.

WHY THE MECHANICS MATTER

A central raised block defeats a lazy floor shot; the bankable roof and sloped return reward accurate power.

INTENDED SOLUTION

1 shot target (unverified): (95,538) -> (270,300) -> (540,240) -> (755,460) -> (828,520) ...

ALTERNATIVE OR EXPLOIT

Two kicks over the block; short bounce off the wedge.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 2 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

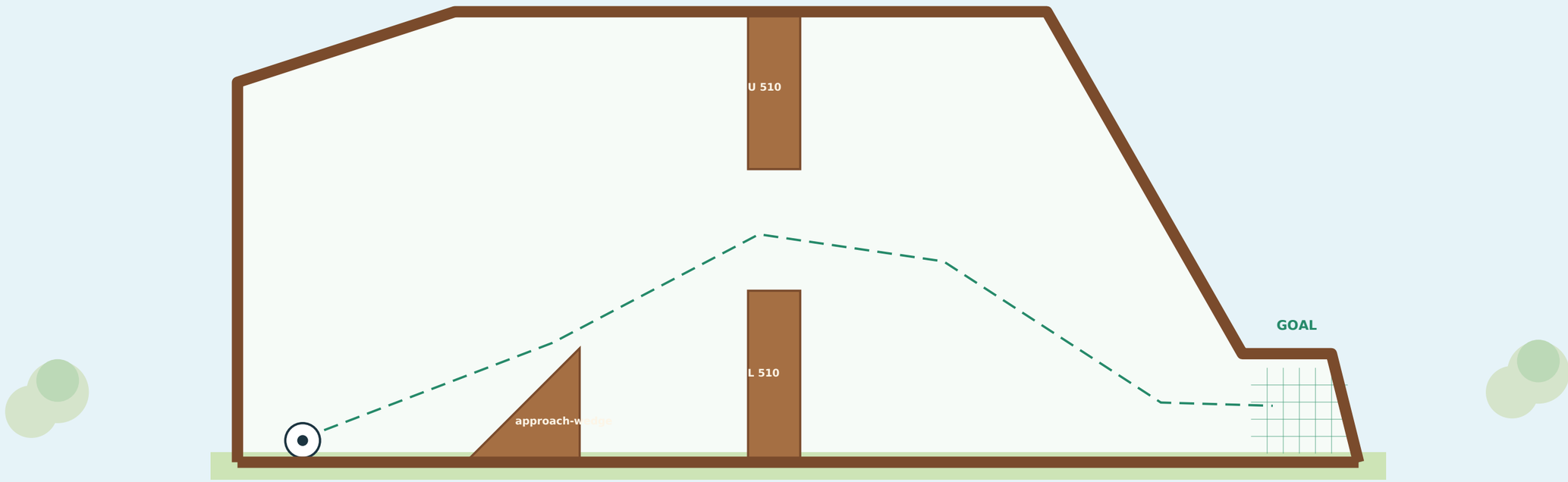
Verify intended bank is achievable at default kick strength; do not advertise a guaranteed one-shot.

42 Window of Opportunity

SINGLE-SCREEN • DIFFICULTY 3/5 • SUGGESTED OPTIMAL 1 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED **PARTITION** **PRECISION**

42 / WINDOW OF OPPORTUNITY



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Shoot through a mid-height gap between two solid dividers.

WHY THE MECHANICS MATTER

A floor-to-roof divider removes all routes except a 112-unit-high midair window.

INTENDED SOLUTION

1 shot target (unverified): (100,538) -> (330,450) -> (520,350) -> (690,375) -> (890,505) ...

ALTERNATIVE OR EXPLOIT

Use a careful second kick near the wedge to line up the window.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 3 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

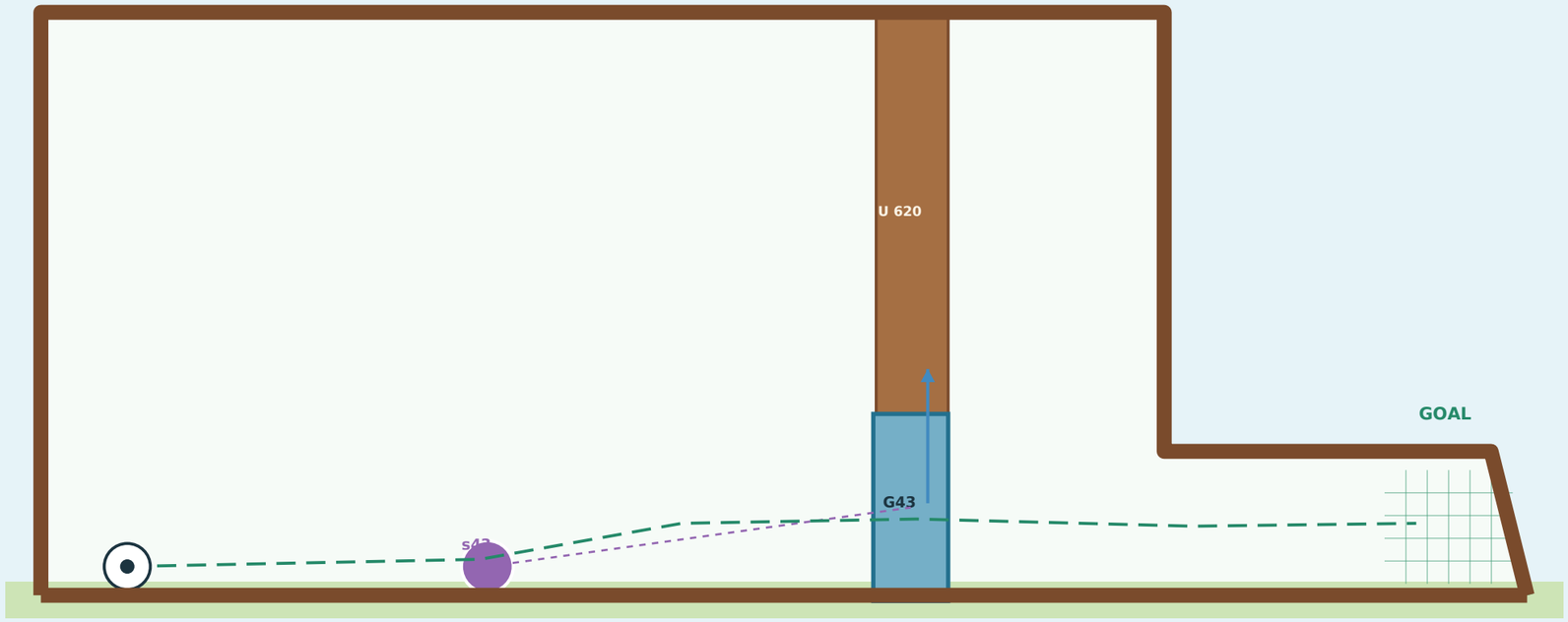
Passage is only 112px high; ball diameter 32 means 80px effective clearance.

43 Unlock the Passage

SINGLE-SCREEN • DIFFICULTY 3/5 • SUGGESTED OPTIMAL 2 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED **SWITCH** **REQUIRED-GATE**

43 / UNLOCK THE PASSAGE



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Hit the switch to open a compulsory gate through an otherwise sealed divider.

WHY THE MECHANICS MATTER

The divider spans roof to floor, so the switch-operated opening is the ONLY route toward the goal.

INTENDED SOLUTION

2 shot target (unverified): (100,540) -> (345,535) -> (485,510) -> (648,507) -> (840,512) ...

ALTERNATIVE OR EXPLOIT

Try a single continuous ground kick that triggers the switch and then enters the opening.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 1 | Camera world view 1050 x 505

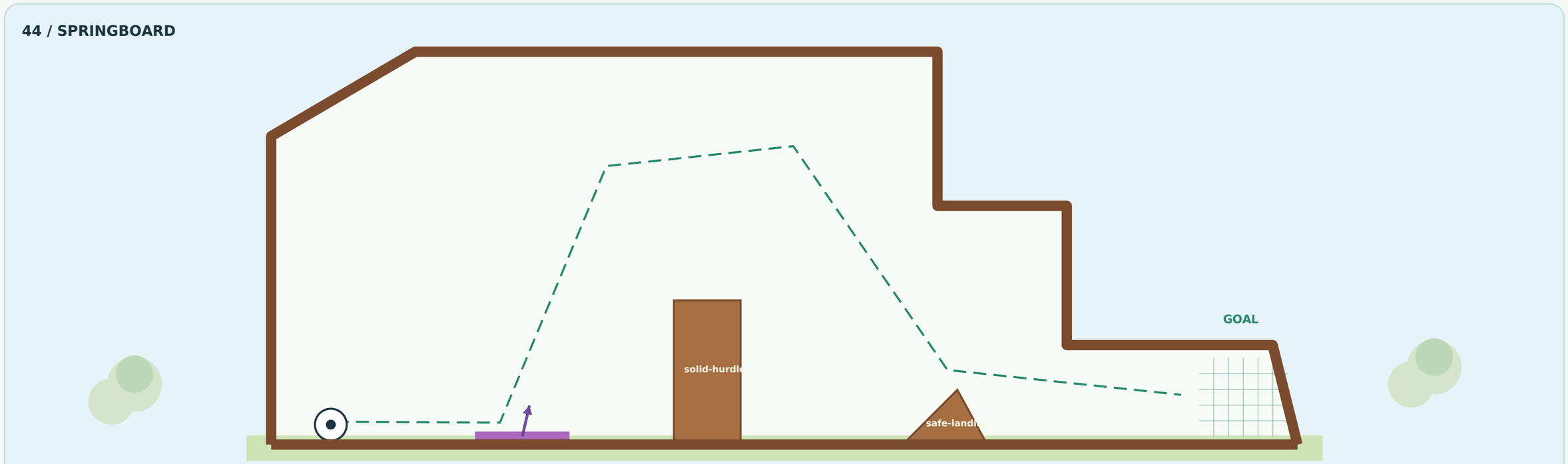
IMPLEMENTATION / RISK CHECK

Confirm that the moving gate clears fully; gate travel must not squeeze the ball between wall colliders.

44 Springboard

SINGLE-SCREEN • DIFFICULTY 2/5 • SUGGESTED OPTIMAL 1 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED | SPRING | ARC



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Use a spring to clear a wall, with a forgiving landing basin.

WHY THE MECHANICS MATTER

Spring is placed before a tall hurdle; an unaided direct roll cannot reach the goal.

INTENDED SOLUTION

1 shot target (unverified): (105,537) -> (270,538) -> (377,280) -> (565,260) -> (720,485) ...

ALTERNATIVE OR EXPLOIT

A high direct kick followed by an airborne correction skips the spring.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 2 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

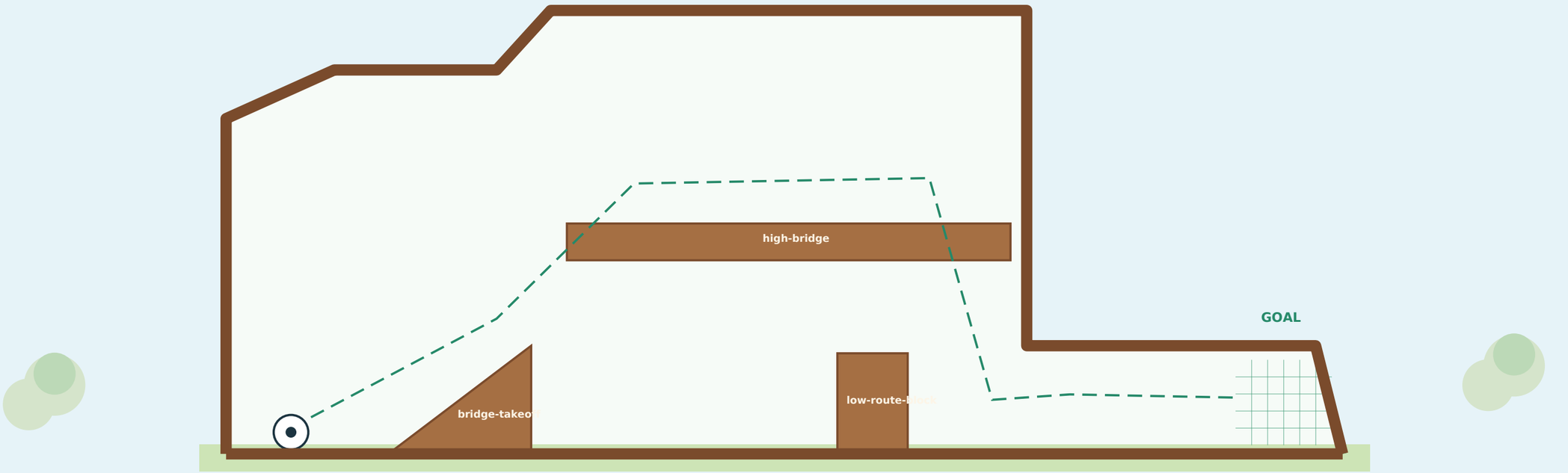
At default spring power, arc must clear the 145px hurdle without striking the low roof.

45 Above or Below

SINGLE-SCREEN • DIFFICULTY 3/5 • SUGGESTED OPTIMAL 1 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED | TWO-ROUTES | RAISED-PLATFORM

45 / ABOVE OR BELOW



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Two distinct routes: a high platform shortcut or a narrow lower passage.

WHY THE MECHANICS MATTER

The overhead bridge avoids the lower solid stop; both routes stay within the enclosed arena.

INTENDED SOLUTION

1 shot target (unverified): (100,536) -> (290,435) -> (417,310) -> (690,305) -> (748,510) ...

ALTERNATIVE OR EXPLOIT

Stay underneath and steer through the lower route with extra shots.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 3 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

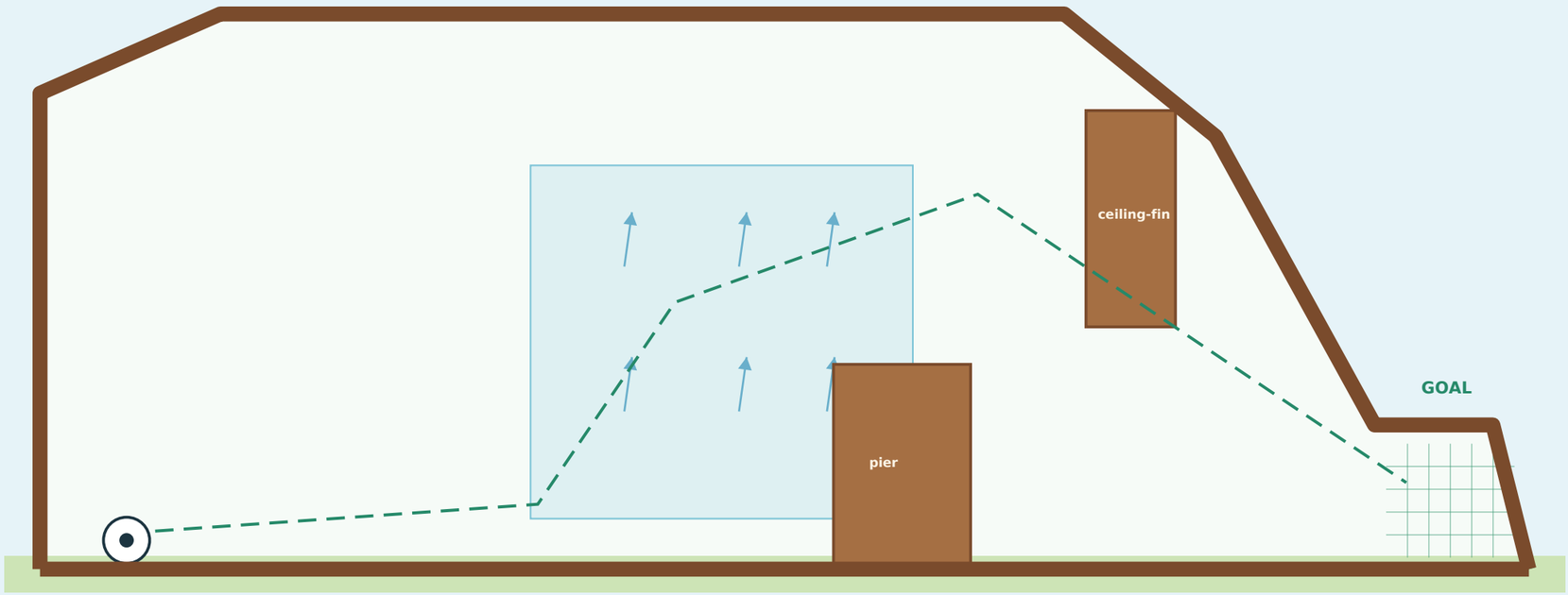
Ensure 34px bridge thickness is visually clear and upper route has clearance.

46 Crosswind Slot

SINGLE-SCREEN • DIFFICULTY 3/5 • SUGGESTED OPTIMAL 2 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED | WIND | AERIAL-ROUTE

46 / CROSSWIND SLOT



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Ride an updraft through a safe opening, then correct the descent.

WHY THE MECHANICS MATTER

Wind supplies lift exactly where the floor obstacle begins; overhead fin makes air control important.

INTENDED SOLUTION

2 shot target (unverified): (100,535) -> (385,515) -> (480,375) -> (690,300) -> (868,420) ...

ALTERNATIVE OR EXPLOIT

Skip the updraft with multiple manual midair kicks.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 2 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

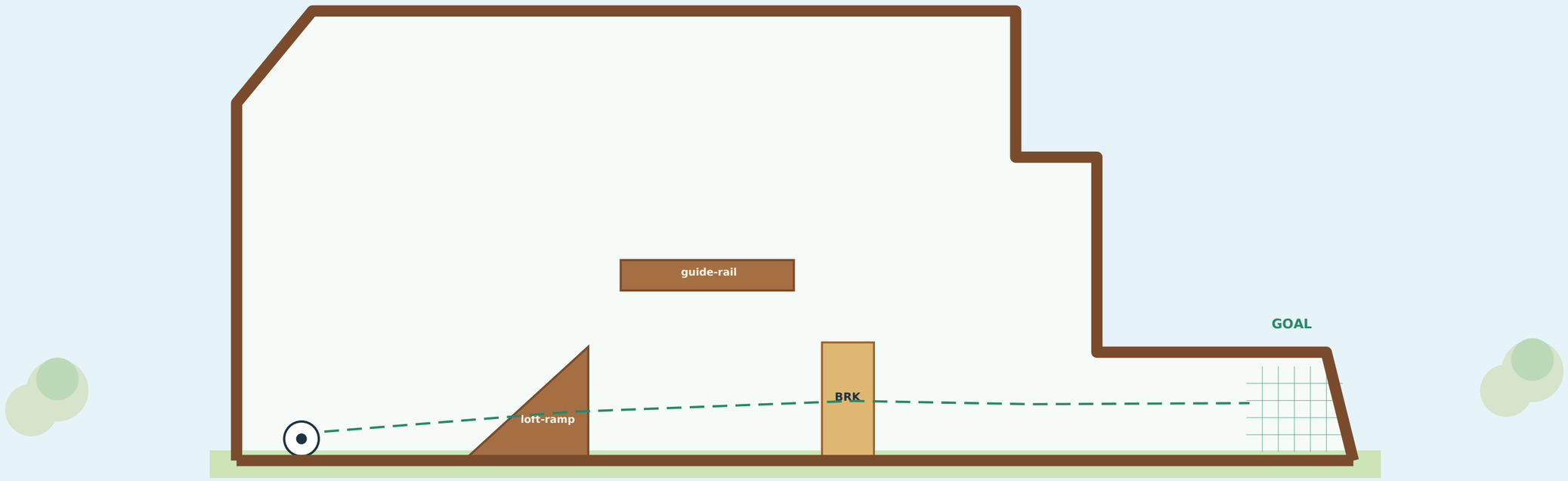
Airflow should be readable, not violently unpredictable.

47 Break or Bypass

SINGLE-SCREEN • DIFFICULTY 3/5 • SUGGESTED OPTIMAL 1 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED **BREAKABLE** **OPTIONAL-SHORTCUT**

47 / BREAK OR BYPASS



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Smash the breakable shortcut, or take an elevated detour.

WHY THE MECHANICS MATTER

The fragile block closes the most direct route, while a loft path remains above it.

INTENDED SOLUTION

1 shot target (unverified): (100,535) -> (350,515) -> (610,505) -> (775,508) -> (976,507) ...

ALTERNATIVE OR EXPLOIT

Launch from the ramp onto the higher rail and bypass the breakable.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 2 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

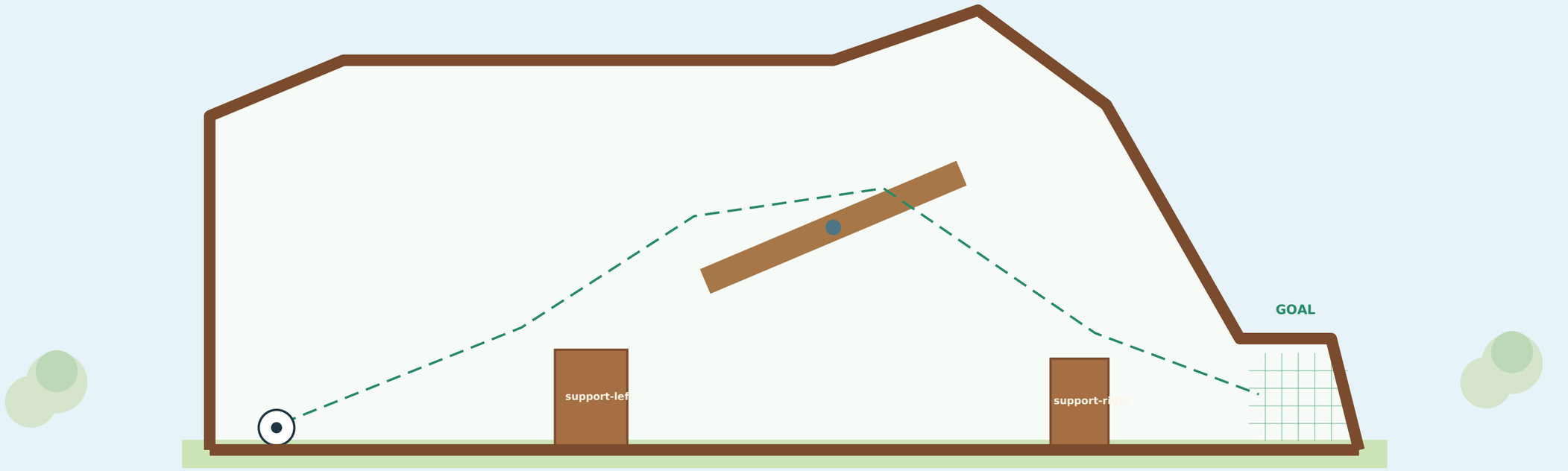
Damage must be achievable at default shot speeds; breakables are not intended to be unbreakable walls.

48 Clockwork Gap

SINGLE-SCREEN • DIFFICULTY 3/5 • SUGGESTED OPTIMAL 1 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED | ROTOR | TIMING

48 / CLOCKWORK GAP



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Wait for a slow rotor to offer an opening, or use it to redirect the ball.

WHY THE MECHANICS MATTER

Rotating blade dominates the central travel corridor and changes the shot timing.

INTENDED SOLUTION

1 shot target (unverified): (105,536) -> (320,450) -> (475,350) -> (645,325) -> (835,455) ...

ALTERNATIVE OR EXPLOIT

Roll low beneath a favourable rotor orientation after a setup kick.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 2 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

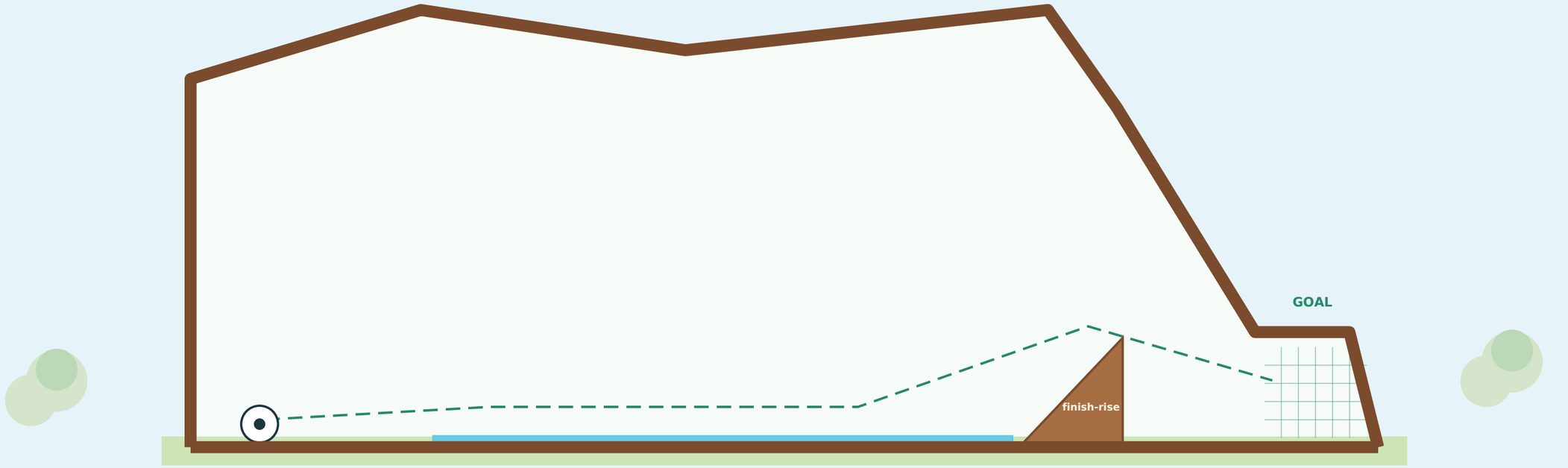
Rotor collision updates must not tunnel or trap the football.

49 Ice Corridor

SINGLE-SCREEN • DIFFICULTY 2/5 • SUGGESTED OPTIMAL 1 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED | ICE | MOMENTUM

49 / ICE CORRIDOR



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Preserve sliding speed across ice then handle a sharp rise.

WHY THE MECHANICS MATTER

A long low-friction runway encourages speed control before a final incline.

INTENDED SOLUTION

1 shot target (unverified): (105,536) -> (300,525) -> (620,525) -> (820,455) -> (980,502) ...

ALTERNATIVE OR EXPLOIT

Short repeated kicks or intentional wall bank.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 1 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

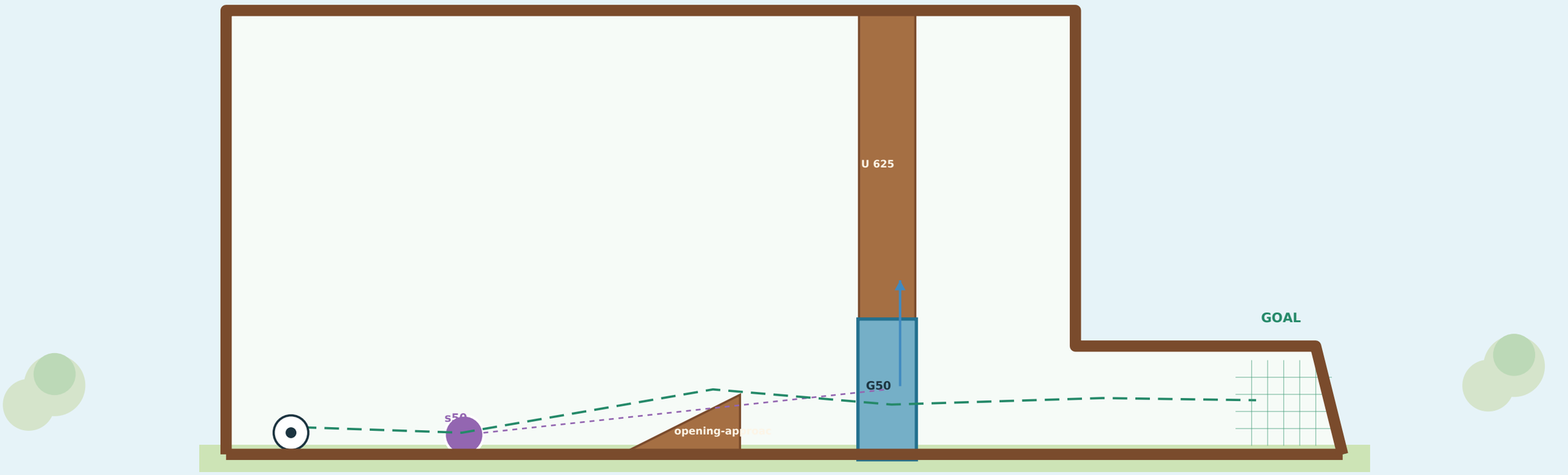
Verify that the low-friction body does not inherit unintended regular-floor friction.

50 The Timed Door

SINGLE-SCREEN • DIFFICULTY 4/5 • SUGGESTED OPTIMAL 2 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED **TIMED-GATE** **SWITCH**

50 / THE TIMED DOOR



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Activate a switch, then reach a compulsory lower gate before it closes.

WHY THE MECHANICS MATTER

The closing gate is essential because the upper partition physically blocks the entire rest of the height.

INTENDED SOLUTION

2 shot target (unverified): (110,535) -> (258,540) -> (490,500) -> (655,514) -> (850,508) ...

ALTERNATIVE OR EXPLOIT

A fast single shot could trigger the timed switch and clear the gate before re-closing.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 2 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

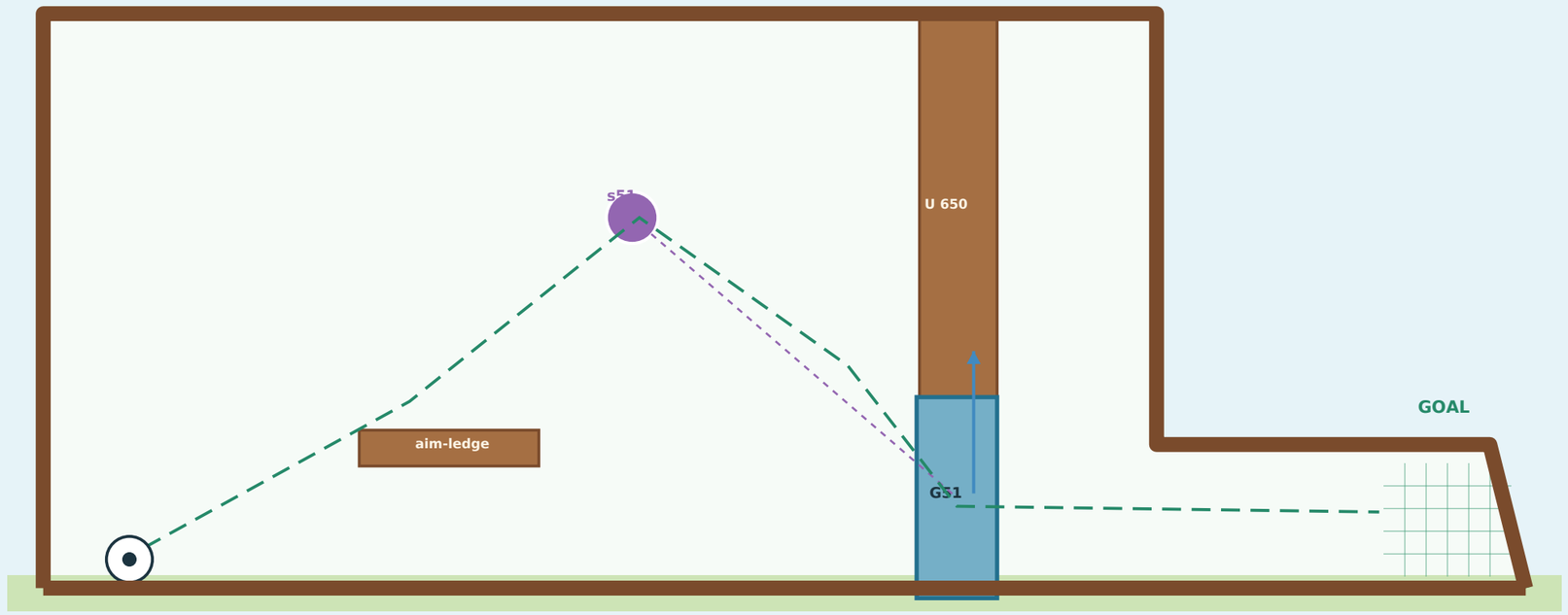
Test switch activation window plus gate travel; 8 seconds must be sufficient for ordinary multi-kick players.

51 High Switch

SINGLE-SCREEN • DIFFICULTY 4/5 • SUGGESTED OPTIMAL 2 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED | SWITCH | HEIGHT-CHANGE | REQUIRED-GATE

51 / HIGH SWITCH



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Trigger an elevated switch to unblock the low ground route.

WHY THE MECHANICS MATTER

Only the gate-side gap connects left and right. An elevated target adds a meaningful preliminary objective.

INTENDED SOLUTION

2 shot target (unverified): (110,532) -> (295,430) -> (455,302) -> (600,405) -> (676,503) ...

ALTERNATIVE OR EXPLOIT

A wall bank can activate the switch in one shot, followed by a low finish.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 2 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

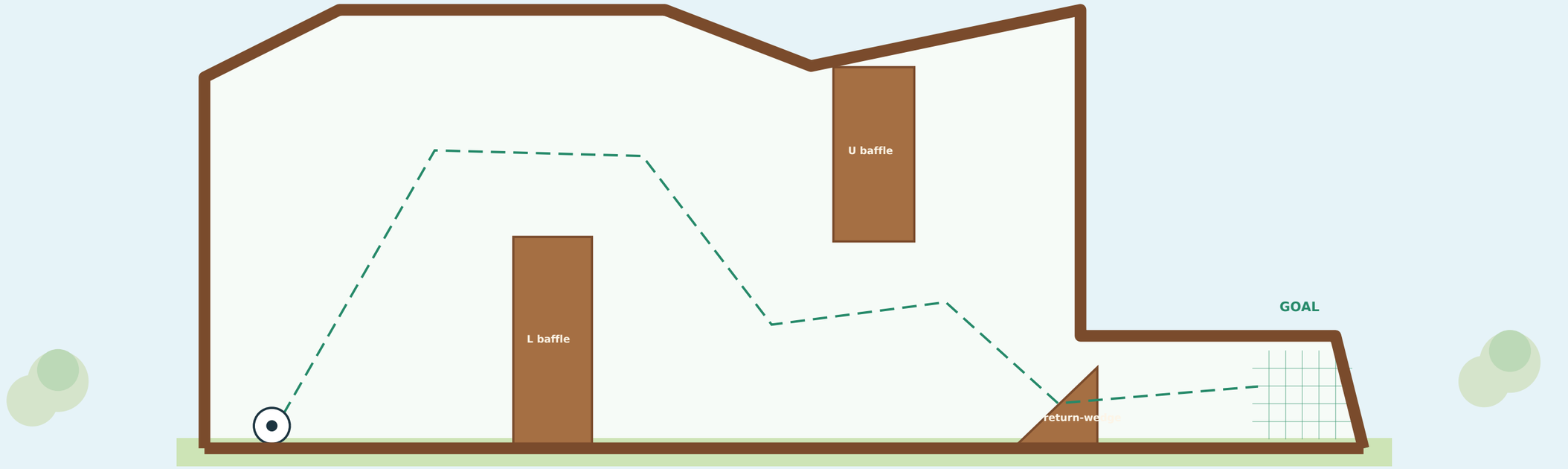
If the upper switch is too high for one kick, reduce its position rather than requiring frame-perfect aim.

52 The Z-Corridor

SINGLE-SCREEN • DIFFICULTY 4/5 • SUGGESTED OPTIMAL 2 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FIXED **ZIGZAG** **BANK**

52 / THE Z-CORRIDOR



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Use staggered upper and lower blocks to thread a non-linear path.

WHY THE MECHANICS MATTER

Staggered floor-to-near-roof blockers remove direct line-of-sight; each kick solves a distinct segment.

INTENDED SOLUTION

2 shot target (unverified): (110,530) -> (245,295) -> (430,300) -> (545,450) -> (700,430) ...

ALTERNATIVE OR EXPLOIT

One-shot double bank for expert players, if physics makes it possible.

Arena 1110 x 595 world units | Start (100, 540) | Right-wall goal (965, 560) | Obstacles 3 | Camera world view 1050 x 505

IMPLEMENTATION / RISK CHECK

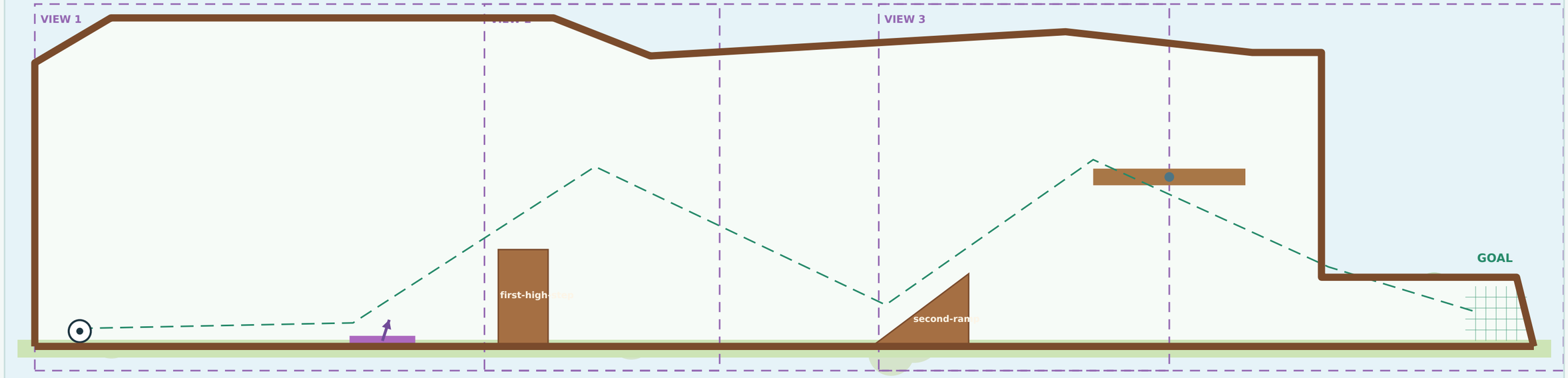
Keep generous gaps around baffle tips: minimum 95px after subtracting ball diameter.

53 Longshot Relay

SCROLLING • DIFFICULTY 3/5 • SUGGESTED OPTIMAL 2 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-X **SCROLL-X** **SPRING** **ROTOR**

53 / LONGSHOT RELAY



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

A rolling course with separated spring and rotor stages.

WHY THE MECHANICS MATTER

Spring is placed just before a tall wall; the rotor then presents a qualitatively different timing challenge.

INTENDED SOLUTION

2 shot target (unverified): (105,594) -> (500,586) -> (850,360) -> (1270,560) -> (1570,350) ...

ALTERNATIVE OR EXPLOIT

Careful kicks can clear each stage without relying entirely on the mechanics.

Arena 2245 x 655 world units | Start (105, 598) | Right-wall goal (2100, 620) | Obstacles 2 | Camera world view 990 x 530

IMPLEMENTATION / RISK CHECK

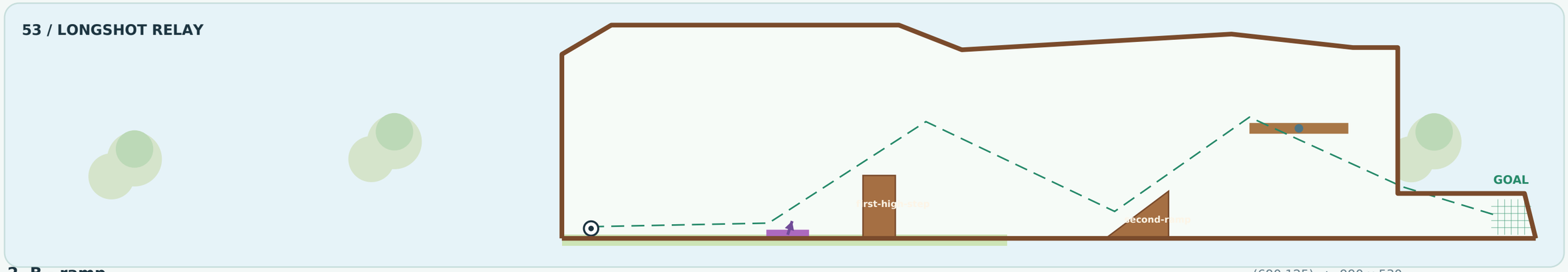
Do not zoom out to show the entire 2,245px world; demonstrate live horizontal camera following.

53 Longshot Relay / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

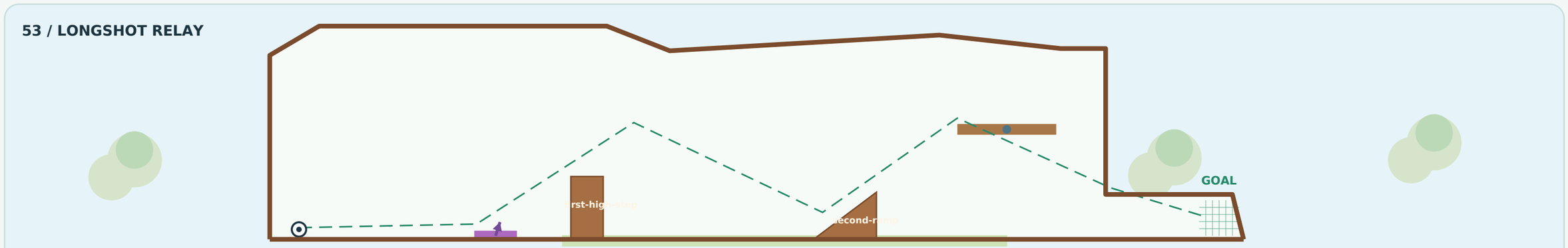
1. A - spring approach

(40,125) + 990 x 530



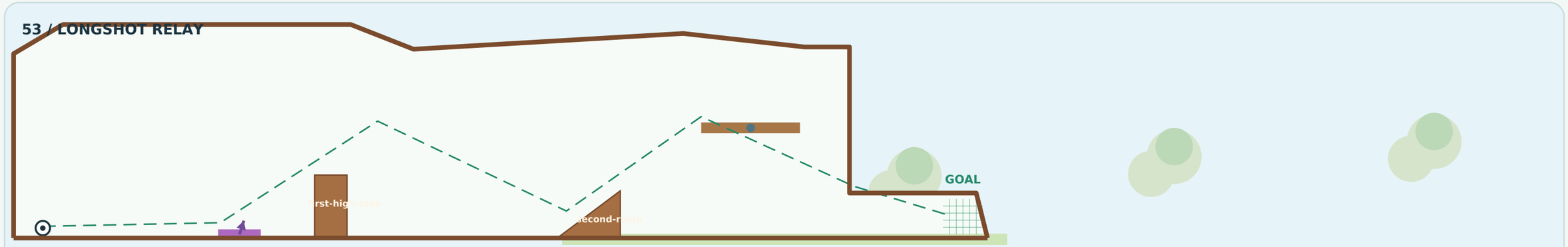
2. B - ramp

(690,125) + 990 x 530



3. C - rotor and goal

(1260,125) + 990 x 530



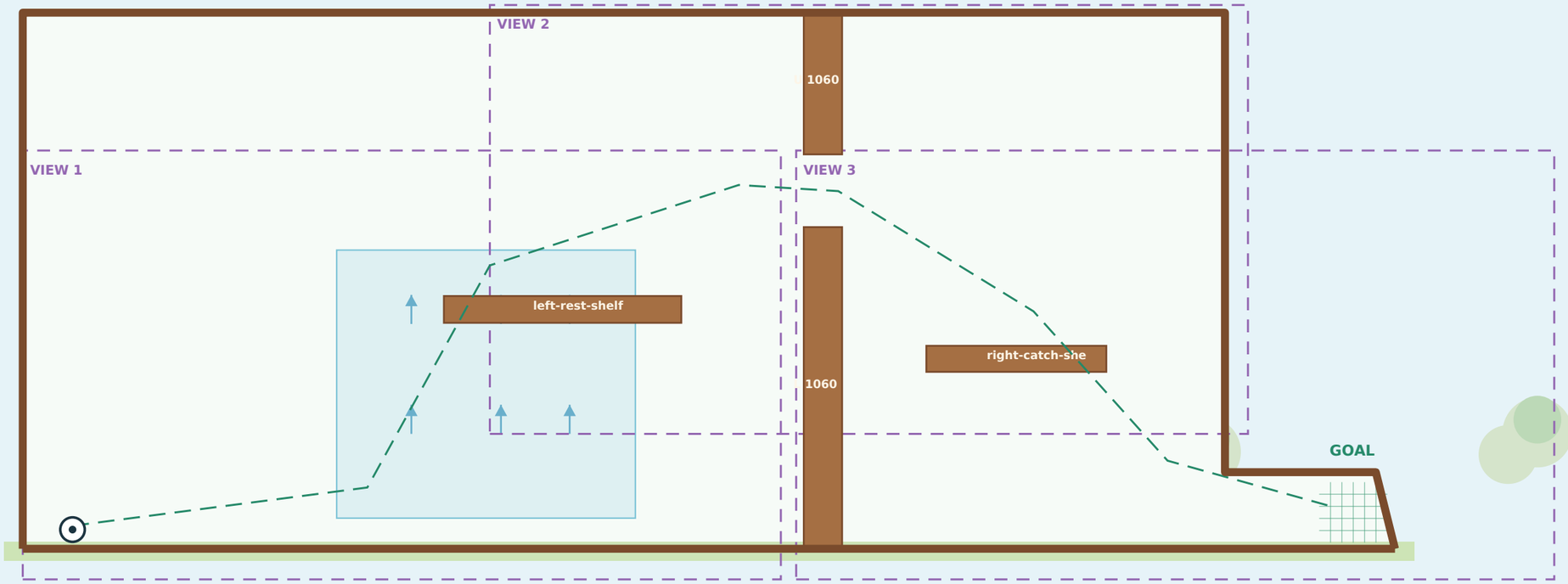
Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

54 Updraft Shaft

SCROLLING • DIFFICULTY 4/5 • SUGGESTED OPTIMAL 3 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-XY **SCROLL-XY** **UPDRAFT** **UPPER-DOORWAY**

54 / UPDRAFT SHAFT



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Climb vertically through an upper doorway, then descend into the goal chamber.

WHY THE MECHANICS MATTER

Full-height divider is traversable only via an elevated window. The wind makes the upper section reachable.

INTENDED SOLUTION

3 shot target (unverified): (105,805) -> (490,755) -> (650,465) -> (975,360) -> (1105,368) ...

ALTERNATIVE OR EXPLOIT

Manual midair kicks or spring-like rebounds could reach the window if strong enough.

Arena 1870 x 870 world units | Start (105, 810) | Right-wall goal (1725, 835) | Obstacles 4 | Camera world view 990 x 560

IMPLEMENTATION / RISK CHECK

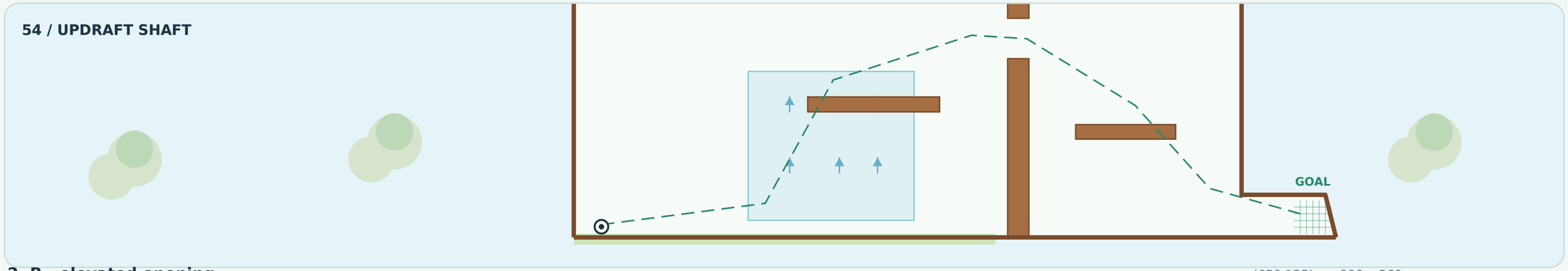
Requires vertical camera tracking and precise view of switch-free high pass; passage 95px tall (63px effective).

54 Updraft Shaft / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

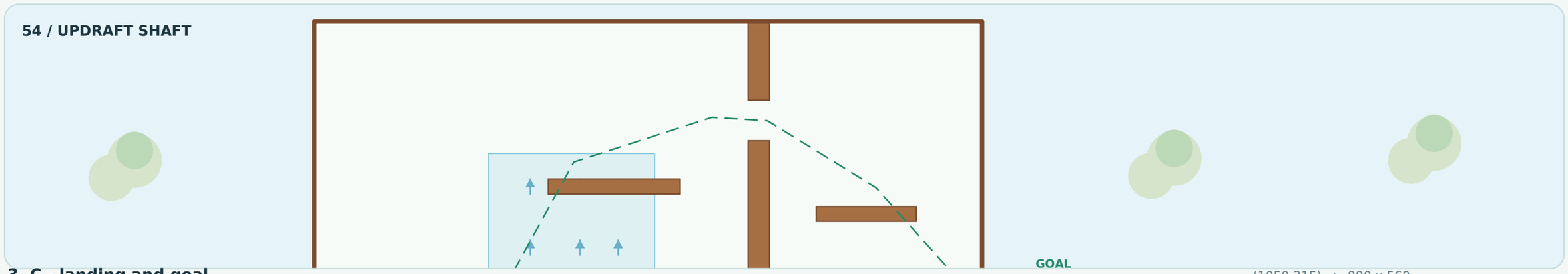
1. A - lower start

(40,315) + 990 x 560



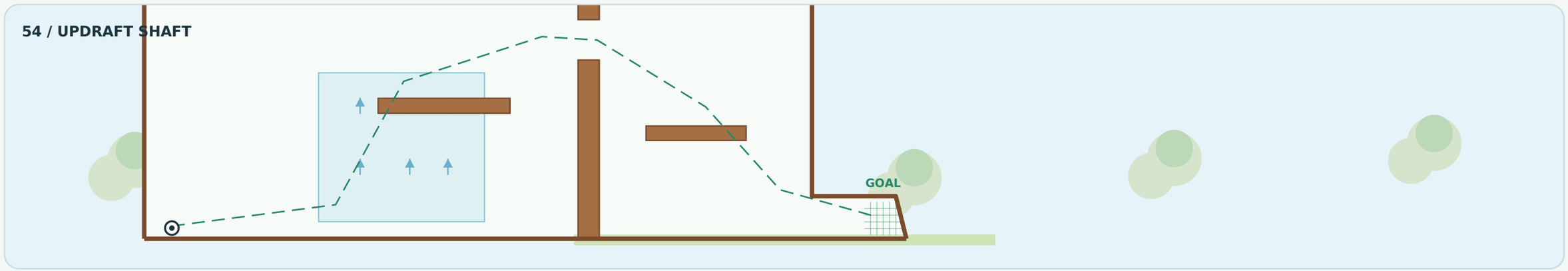
2. B - elevated opening

(650,125) + 990 x 560



3. C - landing and goal

(1050,315) + 990 x 560



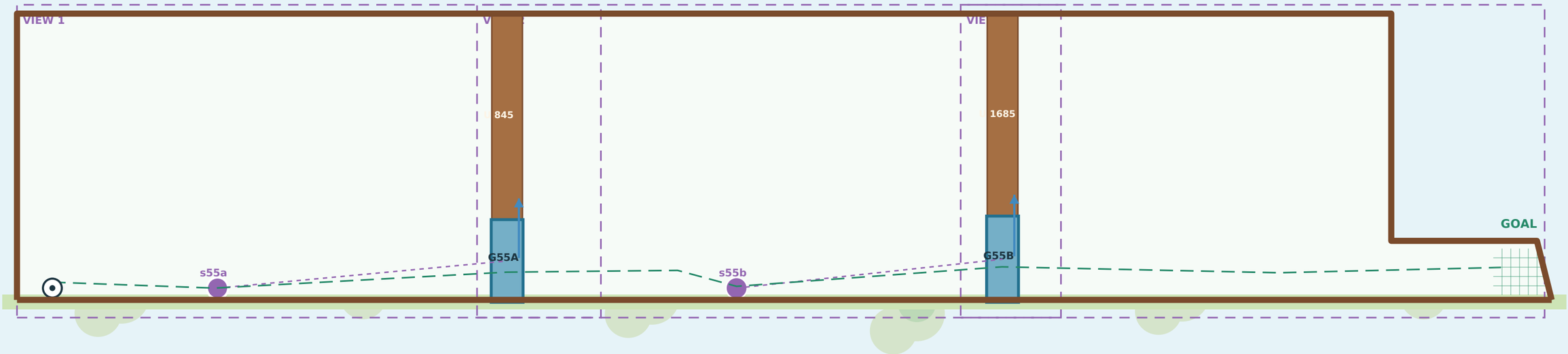
Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

55 Three Chambers

SCROLLING • DIFFICULTY 4/5 • SUGGESTED OPTIMAL 3 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-X **SCROLL-X** **SEQUENTIAL-SWITCHES** **REQUIRED-GATES**

55 / THREE CHAMBERS



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Cross two sealed room dividers by activating a switch in each preceding chamber.

WHY THE MECHANICS MATTER

Two floor-to-ceiling structural partitions provide two genuinely required gated corridors; second switch is past gate one.

INTENDED SOLUTION

3 shot target (unverified): (100,600) -> (375,610) -> (870,583) -> (1160,580) -> (1260,607) ...

ALTERNATIVE OR EXPLOIT

After opening gates, a skilled player may use long rolling shots through each section.

Arena 2680 x 665 world units | Start (100, 610) | Right-wall goal (2535, 630) | Obstacles 2 | Camera world view 990 x 530

IMPLEMENTATION / RISK CHECK

Both switches must only control their intended gates; check clipping at closed-door seams.

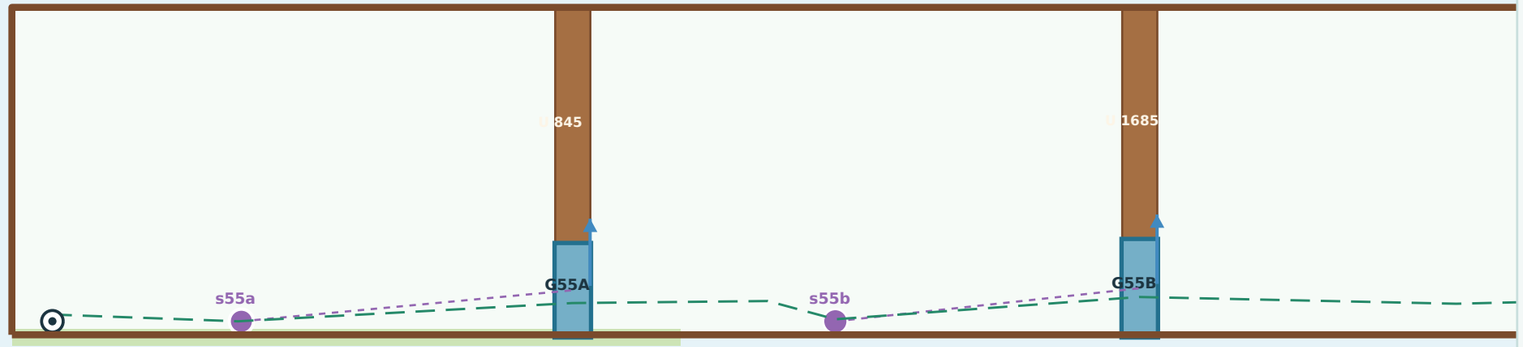
55 Three Chambers / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

1. Room A

(40,130) + 990 x 530

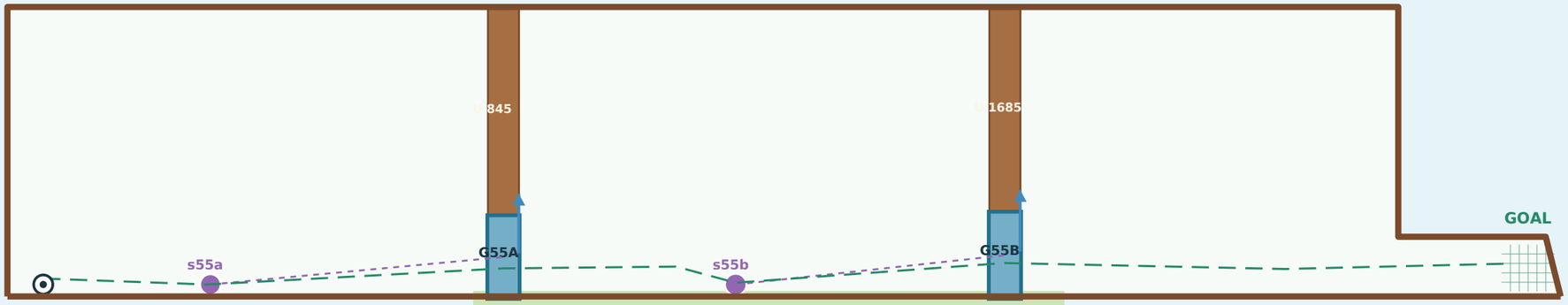
55 / THREE CHAMBERS



2. Room B

(820,130) + 990 x 530

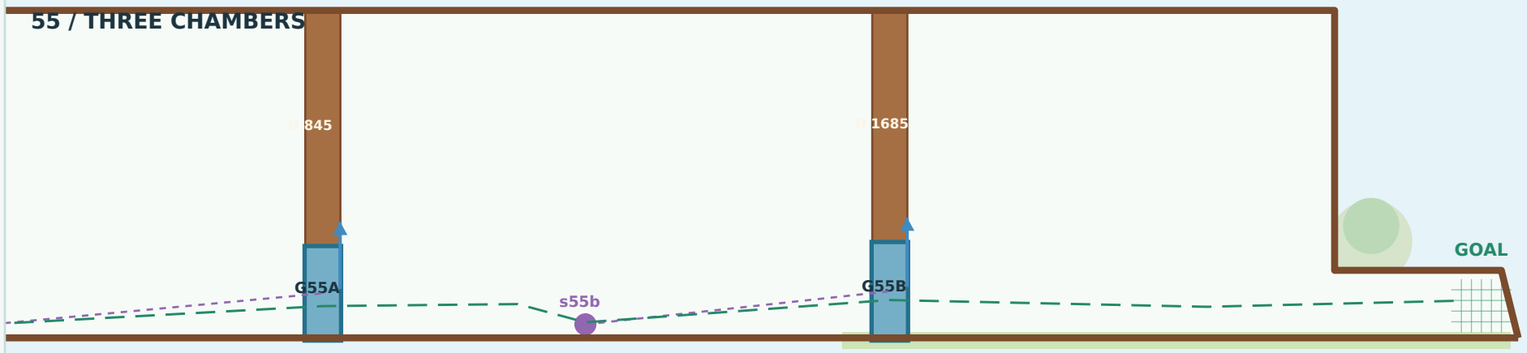
55 / THREE CHAMBERS



3. Room C

(1640,130) + 990 x 530

55 / THREE CHAMBERS

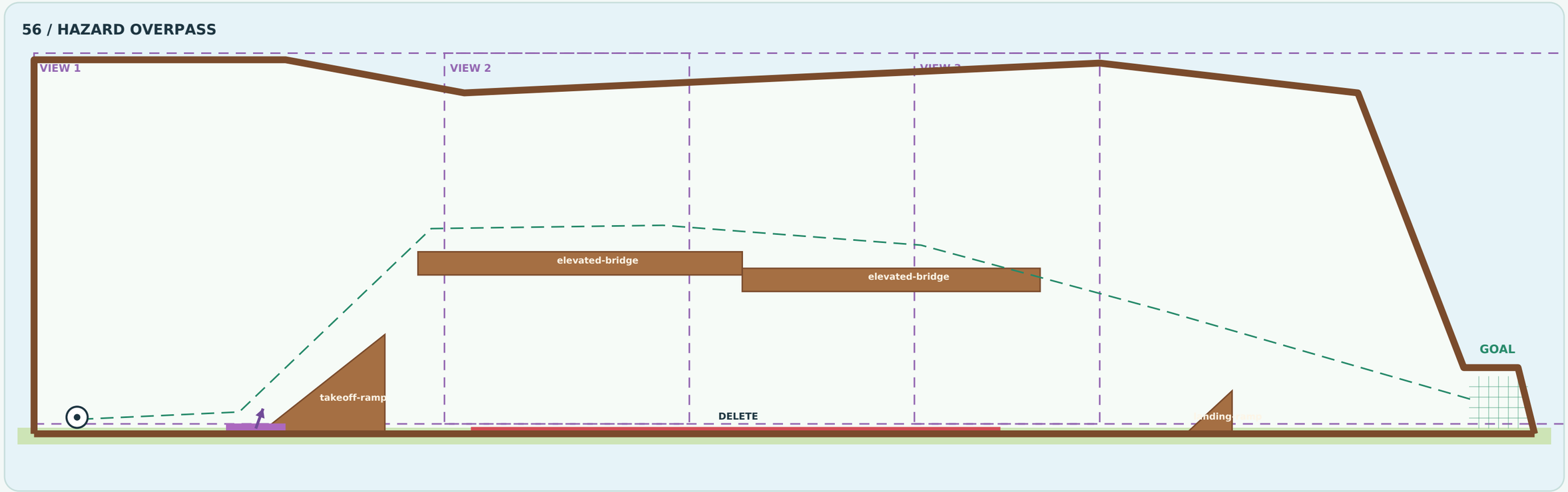


Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

56 Hazard Overpass

SCROLLING • DIFFICULTY 4/5 • SUGGESTED OPTIMAL 2 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-X | SCROLL-X | BRIDGE | HAZARD | SPRING



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Reach an elevated safe route above a long series of red resets.

WHY THE MECHANICS MATTER

Hazard makes pure floor traversal costly; elevated bridge gives a readable safe sequence.

INTENDED SOLUTION

2 shot target (unverified): (110,683) -> (350,672) -> (640,395) -> (990,390) -> (1380,420) ...

ALTERNATIVE OR EXPLOIT

Long aerial shots across the hazard using multiple midair corrections.

Arena 2345 x 740 world units | Start (105, 680) | Right-wall goal (2200, 705) | Obstacles 4 | Camera world view 990 x 560

IMPLEMENTATION / RISK CHECK

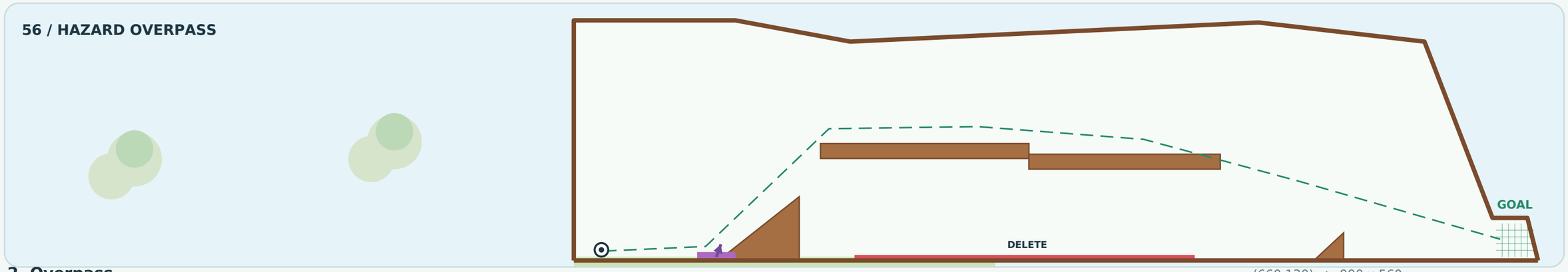
Delete surface must allow 1-shot attempts without accidental unavoidable respawn on the far side.

56 Hazard Overpass / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

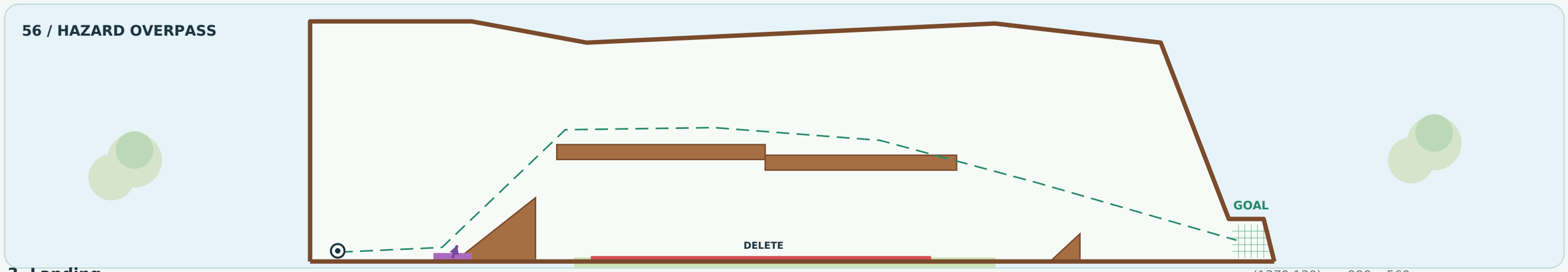
1. Takeoff

(40,130) + 990 x 560



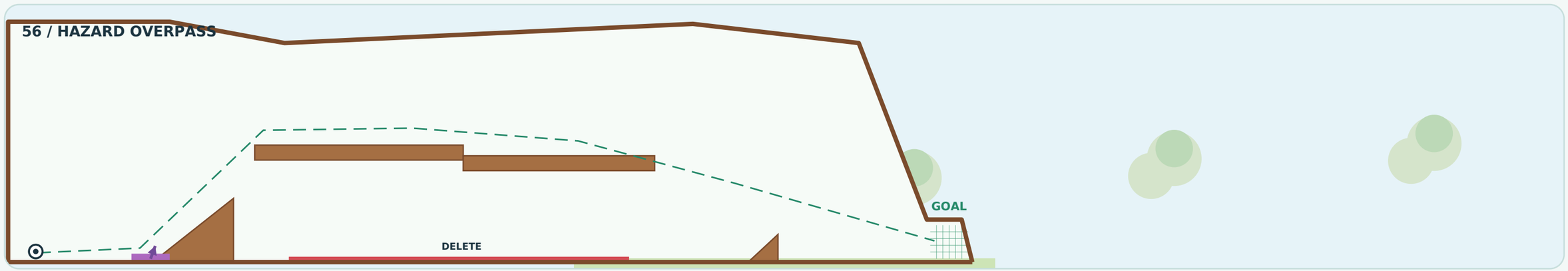
2. Overpass

(660,130) + 990 x 560



3. Landing

(1370,130) + 990 x 560



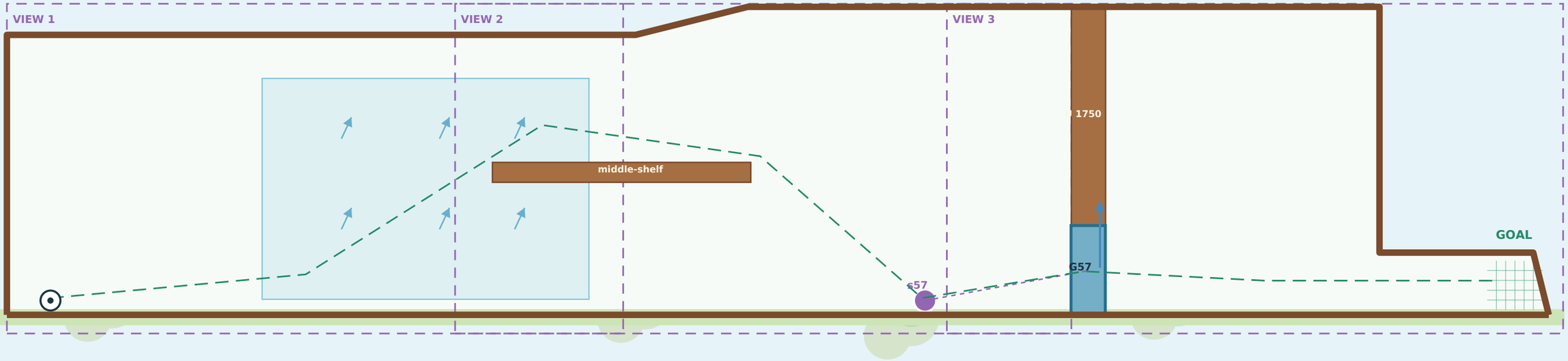
Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

57 Wind and Workshop

SCROLLING • DIFFICULTY 4/5 • SUGGESTED OPTIMAL 3 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-X **SCROLL-X** **WIND** **REQUIRED-GATE**

57 / WIND AND WORKSHOP



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

A wind-assisted passage ending at a mandatory switch-operated door.

WHY THE MECHANICS MATTER

Upward wind changes how to negotiate the raised shelf, but a mandatory gate still controls the last chamber.

INTENDED SOLUTION

3 shot target (unverified): (110,608) -> (520,570) -> (900,330) -> (1250,380) -> (1510,608) ...

ALTERNATIVE OR EXPLOIT

Multi-kick low bypass of some wind effects before compulsory switch.

Arena 2555 x 670 world units | Start (110, 612) | Right-wall goal (2410, 635) | Obstacles 2 | Camera world view 990 x 530

IMPLEMENTATION / RISK CHECK

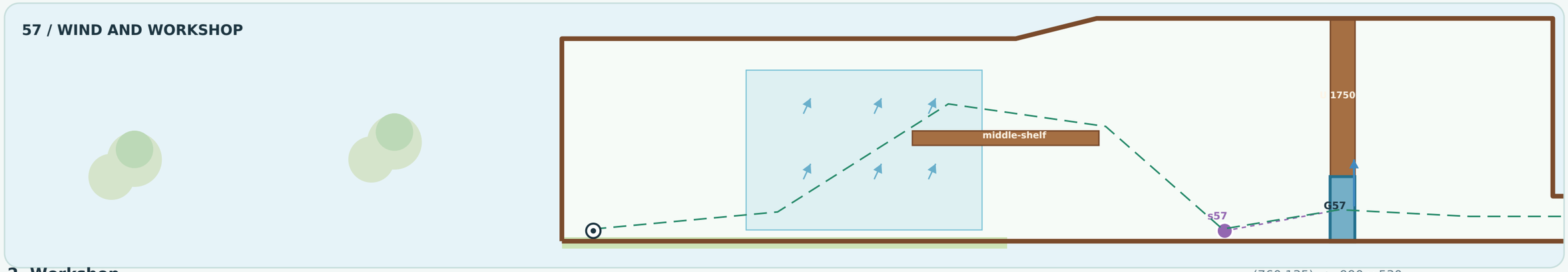
Wind should not throw the ball outside camera view; timing gate remains generous.

57 Wind and Workshop / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

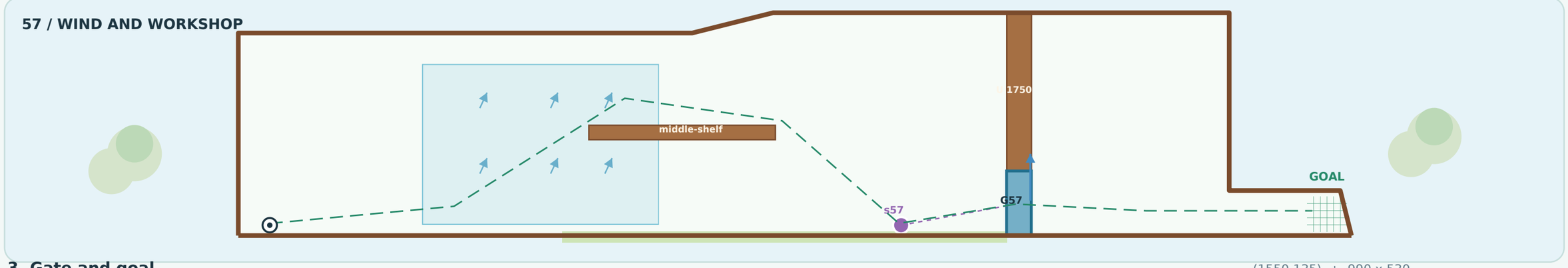
1. Wind gallery

(40,135) + 990 x 530



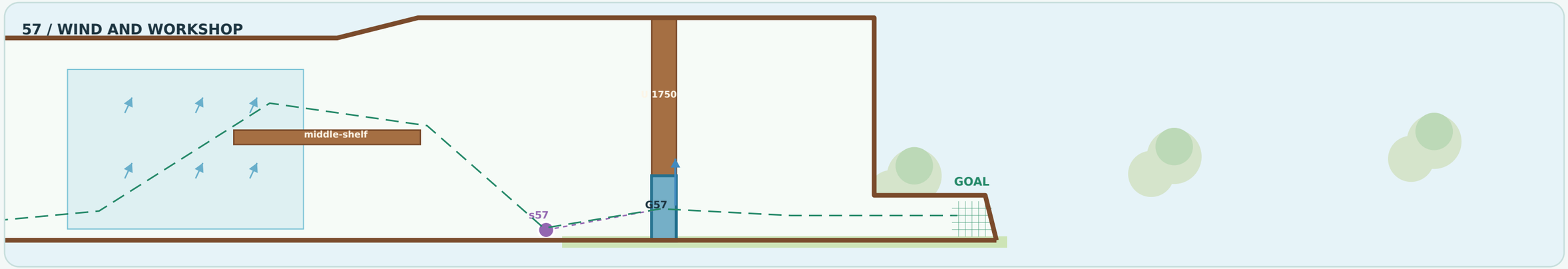
2. Workshop

(760,135) + 990 x 530



3. Gate and goal

(1550,135) + 990 x 530



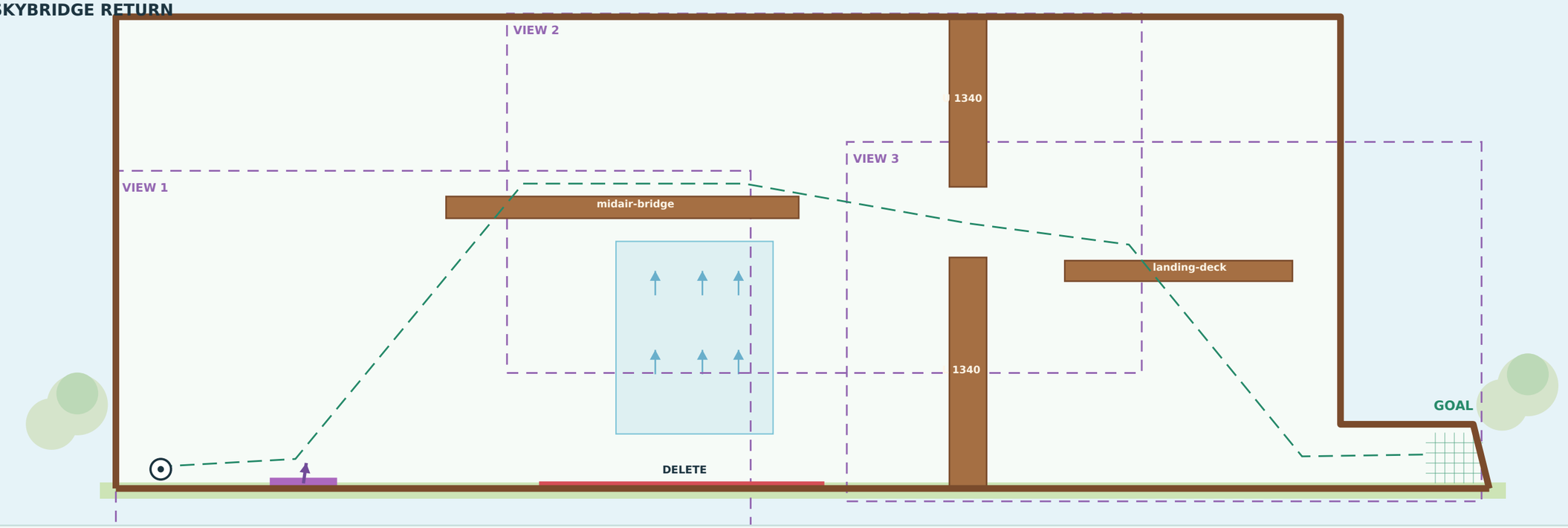
Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

58 Skybridge Return

SCROLLING • DIFFICULTY 5/5 • SUGGESTED OPTIMAL 3 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-XY | SCROLL-XY | HIGH-GATE-GAP | SKYBRIDGE

58 / SKYBRIDGE RETURN



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Rise to the upper bridge; use a narrow midheight doorway to cross.

WHY THE MECHANICS MATTER

Midheight doorway is the only route past a full-height divider, forcing a genuinely multi-area ascent.

INTENDED SOLUTION

3 shot target (unverified): (105,806) -> (320,794) -> (675,365) -> (1020,365) -> (1370,427) ...

ALTERNATIVE OR EXPLOIT

Skilled aerial kicks can bypass the auxiliary wind after reaching bridge height.

Arena 2220 x 875 world units | Start (110, 810) | Right-wall goal (2075, 840) | Obstacles 4 | Camera world view 990 x 560

IMPLEMENTATION / RISK CHECK

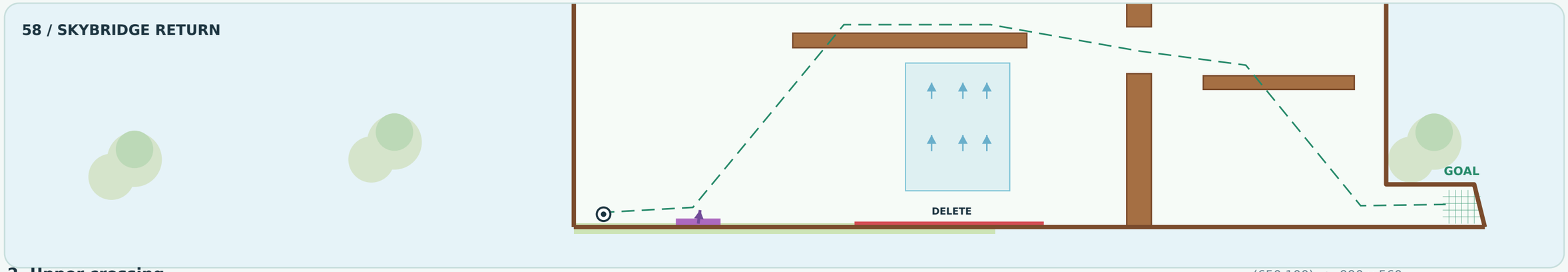
Camera must follow vertical movement without hiding destination platform; safe reset after hazard.

58 Skybridge Return / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

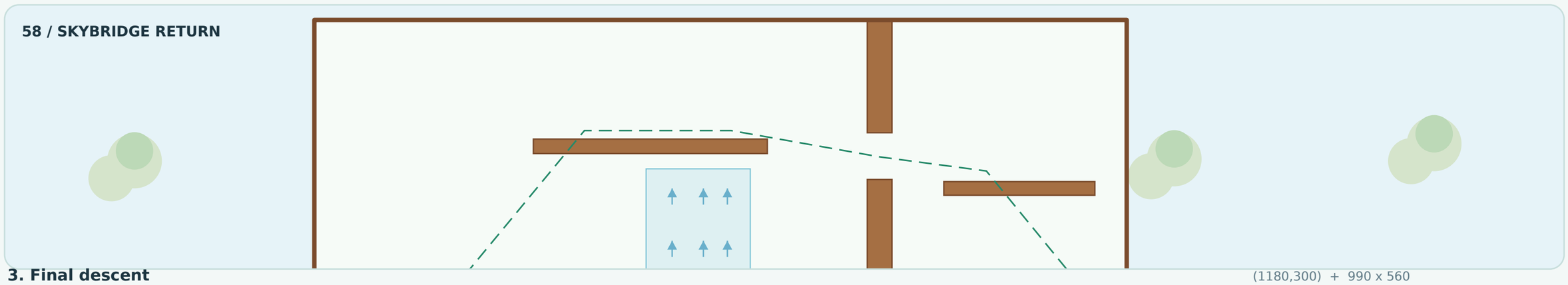
1. Launch basin

(40,345) + 990 x 560



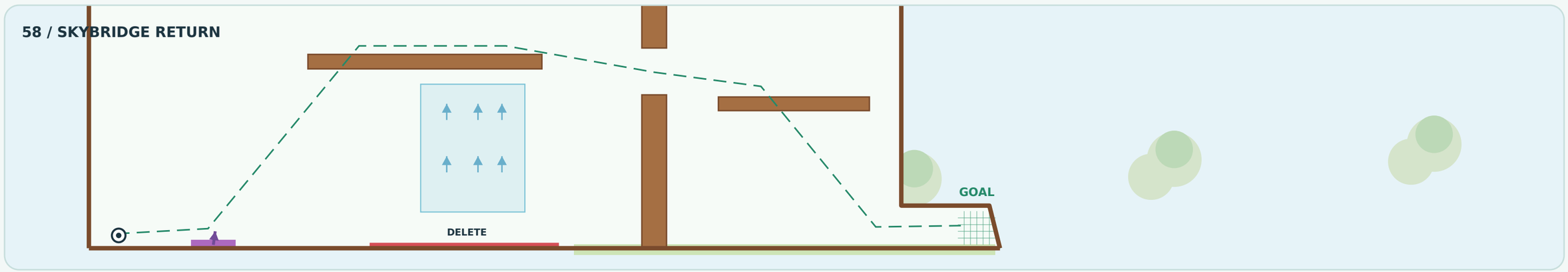
2. Upper crossing

(650,100) + 990 x 560



3. Final descent

(1180,300) + 990 x 560



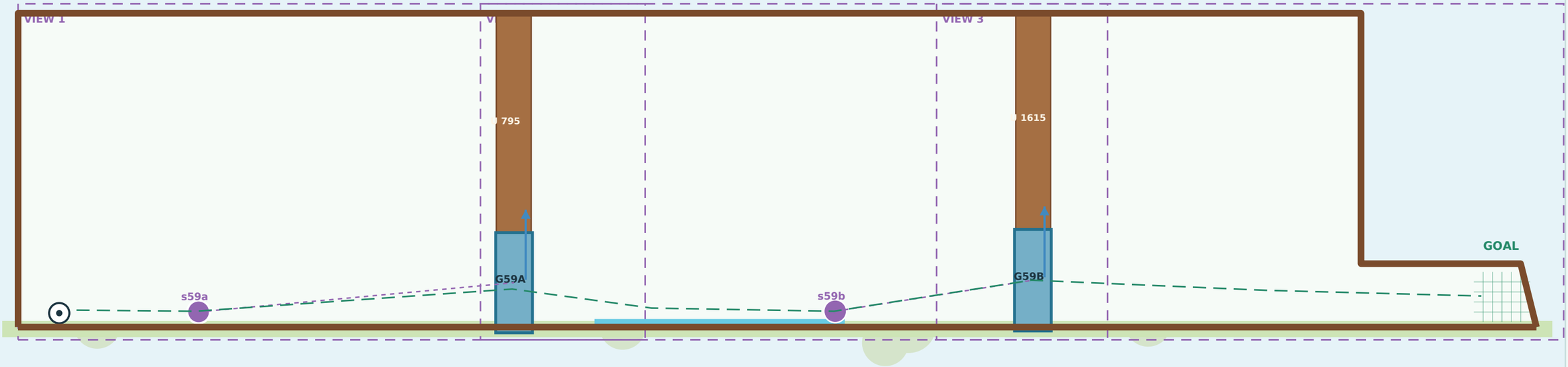
Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

59 Switchback Factory

SCROLLING • DIFFICULTY 5/5 • SUGGESTED OPTIMAL 3 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-X | SCROLL-X | SEQUENTIAL-SWITCHES | ICE

59 / SWITCHBACK FACTORY



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Two meaningful gated partitions and a slippery transition between them.

WHY THE MECHANICS MATTER

A second gate physically separates the last room, and the ice creates a distinct momentum-control segment.

INTENDED SOLUTION

3 shot target (unverified): (100,628) -> (320,630) -> (820,595) -> (1040,625) -> (1328,630) ...

ALTERNATIVE OR EXPLOIT

Time long ground kicks so the ball triggers one switch while already moving towards its gate.

Arena 2475 x 690 world units | Start (105, 633) | Right-wall goal (2330, 655) | Obstacles 2 | Camera world view 990 x 530

IMPLEMENTATION / RISK CHECK

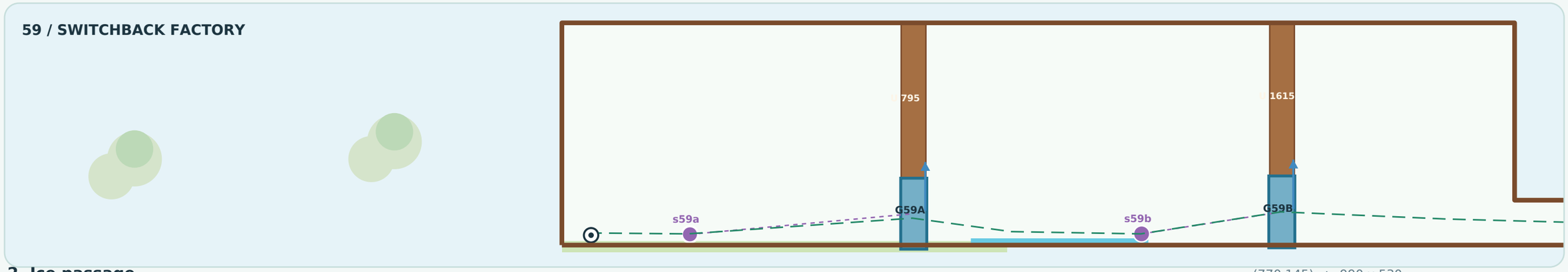
Ice must not send ball into the second closed gate so fast that rebounds appear unpredictable.

59 Switchback Factory / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

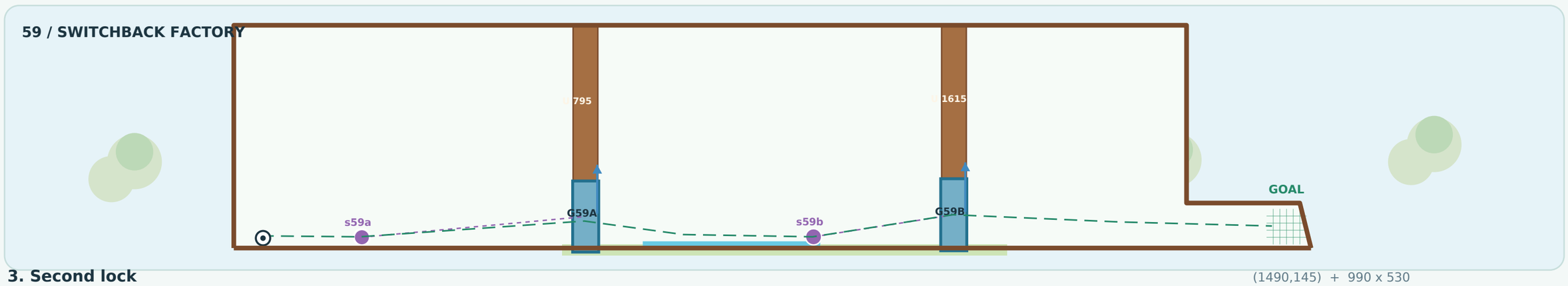
1. First lock

(40,145) + 990 x 530



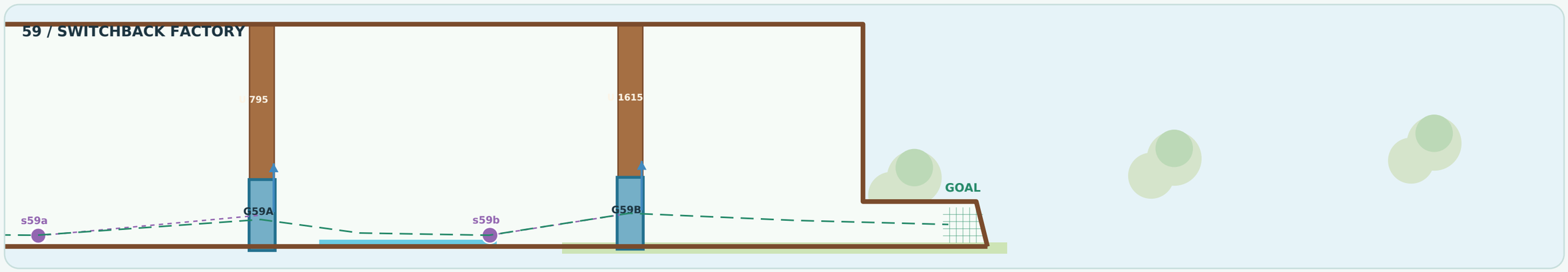
2. Ice passage

(770,145) + 990 x 530



3. Second lock

(1490,145) + 990 x 530



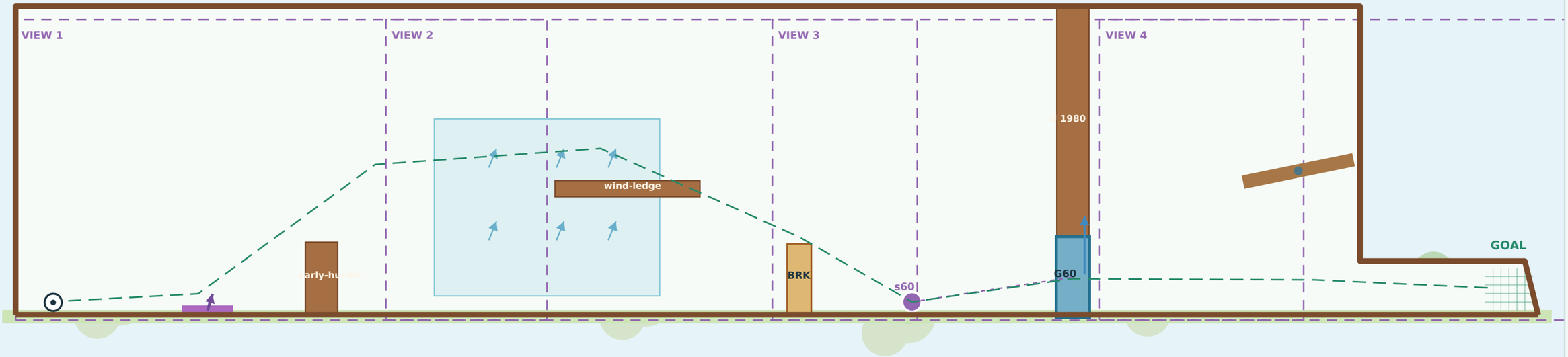
Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

60 The Grand Route

SCROLLING • DIFFICULTY 5/5 • SUGGESTED OPTIMAL 4 SHOT(S) • CONCEPT, NOT PHYSICS-VERIFIED

CAMERA: FOLLOW-X | SCROLL-X | COMBINED | REQUIRED-GATE | FINALE

60 / THE GRAND ROUTE



DASHED GREEN = DESIGNER-INTENDED TRAVEL / PURPLE RECTANGLES = SCROLLING CAMERA VIEWS / NOT A TESTED BALL TRAJECTORY

THE PUZZLE

Five concise puzzle beats: spring, airflow, a breakable shortcut, compulsory gate and finishing rotor.

WHY THE MECHANICS MATTER

Each beat changes the player decision; the final door is compulsory, while spring/wind/wood/rotor allow skillful alternatives.

INTENDED SOLUTION

4 shot target (unverified): (100,706) -> (380,691) -> (710,450) -> (1130,420) -> (1500,585) ...

ALTERNATIVE OR EXPLOIT

Expert multi-kick route can skip the spring and breakable but never bypass the final physical gate.

Arena 2915 x 765 world units | Start (110, 707) | Right-wall goal (2770, 730) | Obstacles 3 | Camera world view 990 x 560

IMPLEMENTATION / RISK CHECK

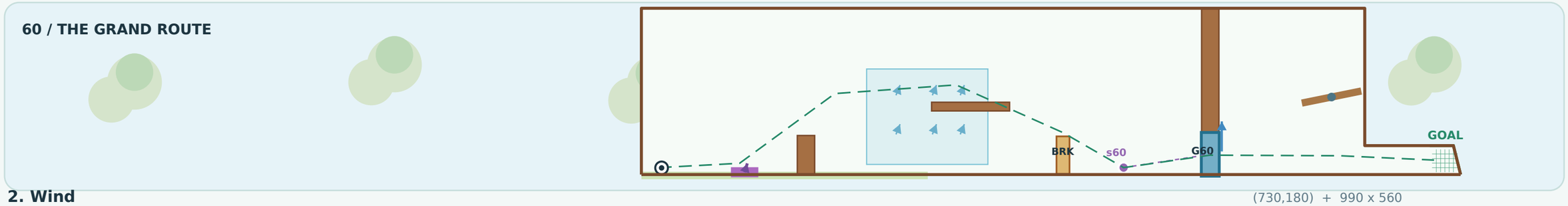
This is an exploratory finale, not a fixed automatic chain; verify camera readability and softlock avoidance section-by-section.

60 The Grand Route / CAMERA REGION DETAILS

INDEPENDENT CROPS FROM THE SAME WORLD COORDINATES • CAMERA MUST FOLLOW BALL, NOT SHRINK FULL LEVEL

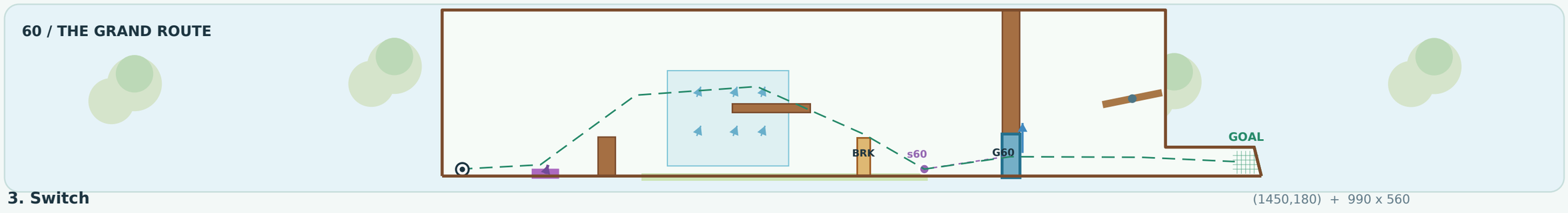
1. Launch

(40,180) + 990 x 560



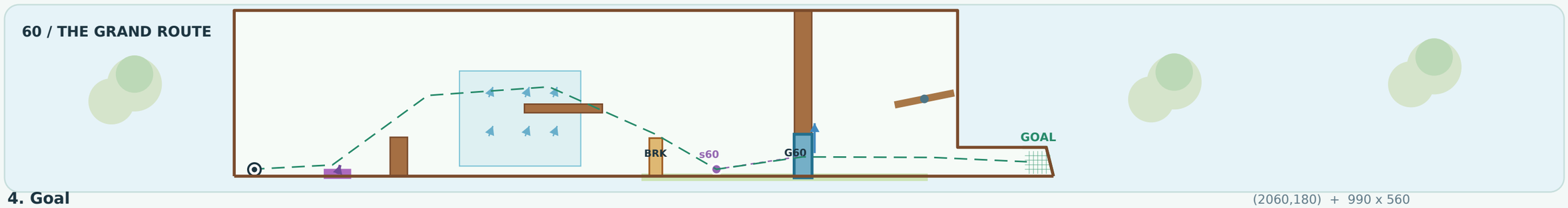
2. Wind

(730,180) + 990 x 560



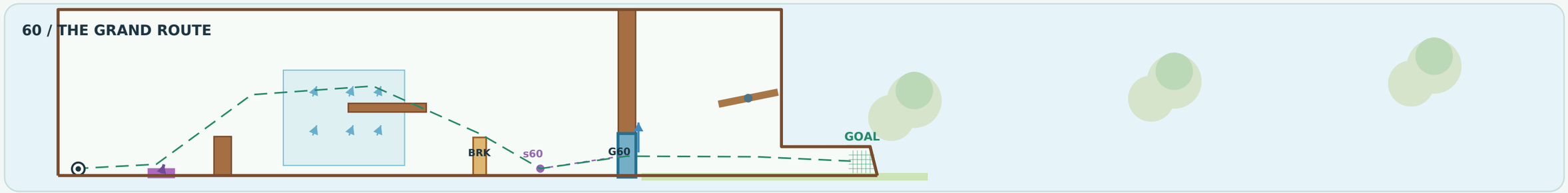
3. Switch

(1450,180) + 990 x 560



4. Goal

(2060,180) + 990 x 560



Camera positions are authored examples. Codex should implement continuous, clamped camera tracking between them, not forced teleport jumps.

CODEX ENGINEERING ROADMAP

IMPLEMENTATION WITHOUT MODIFYING EXISTING LEVELS

1. Add a camera contract

Extend Level with camera?: { mode, viewWidth, viewHeight, deadZone, smoothing, windows? }. Fall back to legacy fixed view for Levels 1-40.

In render.ts, compute visible world rectangle per level; clamp camera to the enclosing map and follow ball within dead zone. Render using camera transforms, not fixed 1250px viewBounds().

Use the SAME camera transform for aiming: screenToWorld and worldToScreen must stay invertible after camera movement and resize.

2. Preserve geometry and mechanics

Generate src/levels-41-60.generated.ts or an imported JSON loader from the canonical JSON data file.

Preserve native Level fields, including outline, goals, surfaces, movers, and mechanics. Keep all obstacles convex; seam-test every enclosing partition.

Test each required gate in CLOSED position; the ball must not pass to the far side through ceilings, floors or gaps. Test full clearance when OPEN.

3. Design acceptance tests

For each level, assert all camera viewports show a ball large enough to aim on a mobile-size landscape viewport.

Play a full physical run using the same player input and physics engine as production. Script successful kicks with trajectory probes; never prove completion by teleporting ball.

Reject levels with pointless mechanisms, impossible switches, unintended escape gaps, unwinnable traps or mandatory frame-perfect timing.

Record shot sequence, goal entry, replay clip/screenshot and stable outcome on repeated deterministic tests.

4. A distinct one-shot celebration

When goal scoring completes and shots === 1, play a separate voice asset announcing GOAL IN ONE!, then a shorter jingle; otherwise use the ordinary goal sound.

Prefer an original recorded or generated voice, not copied original-game audio. Use voice volume + mute setting, web audio unlock and short non-blocking celebration.

Exactly-one-shot must count a single kick even if springs, fans and walls touch the ball repeatedly. Do not block midair kicks and do not introduce shot caps.

ENGINE NOTE: The present goal-mouth test assumes a right-oriented entrance. All designs intentionally retain that orientation.

VALIDATION & HANDOFF

THE STRUCTURE HAS BEEN CHECKED; THE GAMEPLAY HAS NOT

Static validation performed

All 20 unique IDs 41-60; twelve fixed and eight scrolling; consistent right-wall goal placement and dimensions.

All 20 arena enclosure polygons are valid. Start positions lie inside bounds. Obstacles are convex polygons compatible with Matter.js Bodies.fromVertices usage.

All authored gates have linked switches; no loose gate IDs. Goal segments remain open by design.

Not validated in this document

No actual Matter.js simulation/playwright walkthrough of the new levels was run. Suggested optimal shots are targets, not established records.

Spring heights, launch trajectories, break thresholds and gate passage timing still need measured in-engine adjustment.

Mobile visual size assumptions still need tests on representative iPhone landscape viewports.

Codex staged implementation

Stage A: build camera transform and input inversion; run tests for existing Levels 1-40.

Stage B: import fixed Levels 41-52 and playtest each. Resize geometry or adjust object strength if unsuccessful; log changes back to JSON.

Stage C: import scroll Levels 53-60; inspect each camera section and gate continuity.

Stage D: capture screenshots, verify all 60 levels, add one-shot voiceover, and create a gameplay review sheet for user selection.

Machine-readable handoff

Use Goal_In_One_Levels_41-60.json. Canonical native fields match source interfaces; camera and design fields are intentionally new metadata.

Maintain level order and previous saved best shots, add 41-60 as experimental/unlocked for development.

Before shipping: inspect every new stage on phone, verify shots >10, ability to kick in air, and physical ball crossing into goal mouth.

Codex should keep its own list of deviations from authored geometry and explain any change made for physical solvability.

PRIMARY QUALITY BAR: No proposed mechanism should be merely decorative. A switch should matter because a closed, sealed gate prevents progress, not because a description claims it does.